

# Harmonic oscillator 1

Let  $H$  be the Hamiltonian for a harmonic oscillator defined by  $m$  and  $\omega$ .

$$H = -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + \frac{1}{2} m \omega^2 x^2$$

We seek  $\psi(x, t)$  that solves the Schrodinger equation

$$i\hbar \frac{\partial}{\partial t} \psi(x, t) = H \psi(x, t)$$

The solution is

$$\psi_n(x, t) = C_n \exp\left(-\frac{iE_n t}{\hbar}\right) \exp\left(\frac{m\omega x^2}{2\hbar}\right) \frac{d^n}{dx^n} \exp\left(-\frac{m\omega x^2}{\hbar}\right)$$

where  $n = 0, 1, 2, \dots$  and  $C_n$  is the normalization constant

$$C_n = \frac{(-1)^n}{\sqrt{2^n n!}} \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \left(\frac{\hbar}{m\omega}\right)^{n/2}$$

$E_n$  is the energy level

$$E_n = \left(n + \frac{1}{2}\right) \hbar\omega$$

Taking the time derivative of  $\psi_n(x, t)$  we find

$$i\hbar \frac{\partial}{\partial t} \psi_n(x, t) = E_n \psi_n(x, t)$$

Hence by the Schrodinger equation

$$E_n \psi_n(x, t) = H \psi_n(x, t)$$

Factor  $\psi_n(x, t)$  as

$$\psi_n(x, t) = \exp\left(-\frac{iE_n t}{\hbar}\right) \psi_n(x)$$

By substitution

$$E_n \exp\left(-\frac{iE_n t}{\hbar}\right) \psi_n(x) = H \left[ \exp\left(-\frac{iE_n t}{\hbar}\right) \psi_n(x) \right]$$

Since  $H$  is independent of  $t$  we can factor the right-hand side as

$$E_n \exp\left(-\frac{iE_n t}{\hbar}\right) \psi_n(x) = \exp\left(-\frac{iE_n t}{\hbar}\right) H \psi_n(x)$$

The exponentials cancel leaving the time-independent Schrodinger equation

$$E_n \psi_n(x) = H \psi_n(x)$$

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